

SOFTWARE TESTER (QA) ASSESSMENT

Task Management Application – Test Case Design & Potential Risk Analysis

Field	Details
Candidate Name	Munna Kumar
Email ID	munna.kumar.cs.2023@mitmeerut.ac.in
College	Meerut Institute of Technology, Meerut
Assessment Focus	Functional Testing, Validation, Error Handling & Risk Identification

Assessment Overview

This assessment presents a structured QA approach for a Task Management Application. The application supports user registration and login, and logged-in users can create, view, edit, and delete tasks. Tasks are stored in a database and displayed in a list.

The test design includes positive, negative, and edge cases, followed by potential bugs and risk areas identified without executing the application.

Note: The potential bugs listed in this document are hypothetical risks and should be verified through actual execution before being reported as confirmed defects.

1. Testing Objectives

- Design clear test scenarios and test cases for Registration, Login, and Task CRUD operations.
- Cover positive, negative, and edge cases.
- Verify input validation and error handling.
- Consider authentication, authorization, data persistence, and data integrity risks.
- Identify 8–10 potential bugs or risk areas and assign severity based on impact.

2. Application Scope

Area	Functionality
Authentication	User Registration and Login
Task Management	Create, View/List, Edit, and Delete tasks
Data	Tasks are stored in a database and displayed in a list
Validation	Required fields, invalid input, boundary and unexpected input
Error Handling	Invalid requests, server/database failures, and user-facing error messages

3. Test Case Design

The following test cases use the format: **Test Case ID, Scenario, Steps/Test Data, Expected Result, Type.**

A. Registration

ID	Scenario	Steps / Test Data	Expected Result	Type
REG-01	Valid registration	Enter valid name, email and password; submit.	Account is created successfully.	Positive
REG-02	Empty name	Leave name blank; submit.	Required-field validation is shown.	Negative
REG-03	Empty email	Leave email blank; submit.	Email required message is shown.	Negative
REG-04	Empty password	Leave password blank; submit.	Password required message is shown.	Negative
REG-05	Invalid email	Enter abc@ and submit.	Invalid email validation is shown.	Negative
REG-06	Existing email	Register using an already registered email.	Registration is rejected with a clear message.	Negative
REG-07	Weak password	Enter a password below the allowed strength/length.	Password validation is shown.	Negative
REG-08	Password mismatch	Enter different password and confirmation values, if confirmation exists.	Registration is not completed.	Negative
REG-09	Leading/trailing spaces	Enter spaces around valid name/email.	Input is trimmed or handled consistently.	Edge
REG-10	Maximum-length input	Enter data at maximum supported length.	System accepts valid boundary input without error.	Edge
REG-11	Extremely long input	Enter values beyond expected maximum.	Input is rejected or safely handled.	Edge

B. Login

ID	Scenario	Steps / Test Data	Expected Result	Type
LOG-01	Valid login	Enter registered email and correct password.	User logs in successfully.	Positive
LOG-02	Wrong password	Enter valid email and incorrect password.	Login fails with an appropriate error.	Negative
LOG-03	Unregistered email	Enter an email not present in the system.	Login fails without exposing sensitive details.	Negative
LOG-04	Empty email	Leave email blank; submit.	Email validation is shown.	Negative
LOG-05	Empty password	Leave password blank; submit.	Password validation is shown.	Negative
LOG-06	Invalid email format	Enter invalid email syntax.	Validation message is displayed.	Negative
LOG-07	Both fields empty	Submit the form without input.	Required-field errors are shown.	Negative
LOG-08	Password case sensitivity	Change password capitalization.	Authentication behaves according to password rules; incorrect value is rejected.	Edge
LOG-09	Email with spaces	Use leading/trailing spaces around email.	Input is handled consistently and securely.	Edge
LOG-10	Repeated failed attempts	Attempt login repeatedly with wrong credentials.	Failures are handled securely without application instability.	Edge

C. Create Task

ID	Scenario	Steps / Test Data	Expected Result	Type
CRT-01	Create with valid data	Enter valid task information and save.	Task is created successfully.	Positive
CRT-02	Required fields only	Provide all mandatory fields and save.	Task is created.	Positive
CRT-03	Empty task title	Leave required title blank.	Validation is shown; task is not created.	Negative
CRT-04	Spaces-only title	Enter only spaces as title.	Input is treated as invalid.	Negative
CRT-05	Optional description empty	Leave description empty, if optional.	Task is created successfully.	Edge
CRT-06	Very long title	Enter title beyond supported length.	Application validates or safely handles input.	Edge
CRT-07	Special characters	Enter supported special characters.	Application handles input safely and consistently.	Edge
CRT-08	Duplicate task	Create the same task twice.	Application behaves according to defined duplicate-handling rules.	Edge
CRT-09	Persistence after refresh	Create task and refresh page.	Created task remains available.	Positive
CRT-10	Create without login	Attempt to access task creation while unauthenticated.	Access is denied or user is redirected to login.	Negative

D. View Task List

ID	Scenario	Steps / Test Data	Expected Result	Type
VIEW-01	View existing tasks	Login and open task list.	User's tasks are displayed correctly.	Positive
VIEW-02	View task details	Open/select an existing task.	Correct task information is displayed.	Positive
VIEW-03	Empty task list	Login as a user with no tasks.	Appropriate empty-state message is displayed.	Edge
VIEW-04	Multiple tasks	Create several tasks and open list.	All relevant tasks are displayed correctly.	Positive
VIEW-05	Refresh list	Refresh the task list page.	Persisted tasks remain available.	Positive
VIEW-06	Unauthorized task access	Try accessing another user's task directly.	Access is denied.	Negative
VIEW-07	After deletion	Delete a task and refresh list.	Deleted task does not reappear.	Positive

E. Edit Task

ID	Scenario	Steps / Test Data	Expected Result	Type
EDIT-01	Edit existing task	Change task information and save.	Updated information is displayed.	Positive
EDIT-02	Edit title	Modify title and save.	New title is persisted.	Positive
EDIT-03	Edit description	Modify description and save.	New description is persisted.	Positive
EDIT-04	Clear required field	Remove required title and save.	Validation error is shown.	Negative
EDIT-05	Invalid data	Enter invalid task data.	Invalid input is rejected.	Negative
EDIT-06	Edit another user's task	Attempt modification without ownership/permission.	Modification is denied.	Negative
EDIT-07	Refresh after update	Update task and refresh page.	Updated data remains saved.	Positive
EDIT-08	Cancel edit	Make changes and cancel, if supported.	Original data remains unchanged.	Edge

F. Delete Task

ID	Scenario	Steps / Test Data	Expected Result	Type
DEL-01	Delete existing task	Select task and confirm deletion.	Task is deleted successfully.	Positive
DEL-02	Cancel deletion	Start deletion and cancel confirmation.	Task remains unchanged.	Positive
DEL-03	Confirm deletion	Confirm delete action.	Task is removed from list.	Positive
DEL-04	Delete non-existing task	Attempt deletion using stale/already deleted task.	Request is safely handled.	Edge
DEL-05	Delete another user's task	Attempt unauthorized deletion.	Deletion is denied.	Negative
DEL-06	Refresh after deletion	Delete task and refresh.	Deleted task does not return.	Positive
DEL-07	Delete last task	Delete the user's only task.	Correct empty-state is displayed.	Edge

G. Input Validation & Error Handling

ID	Scenario	Steps / Test Data	Expected Result	Type
VAL-01	All fields empty	Submit forms without entering required values.	Required-field errors are displayed.	Negative
VAL-02	Invalid email	Enter invalid email format.	Clear validation message is shown.	Negative
VAL-03	Extremely long input	Enter unusually long input.	Application safely rejects or handles it.	Edge
VAL-04	Special characters	Enter special characters in inputs.	Input is safely processed.	Edge
VAL-05	Spaces-only input	Enter only spaces in required fields.	Input is considered invalid.	Negative
VAL-06	Server/database failure	Trigger or simulate a backend failure.	Meaningful user-friendly error is shown; application does not crash.	Negative
VAL-07	Invalid task ID	Request a task using invalid/manipulated ID.	Application handles request safely.	Negative
VAL-08	Unauthorized request	Attempt protected action without permission.	Access is denied.	Negative
VAL-09	Network unavailable	Perform operation while service/network is unavailable.	Suitable error is shown and app remains stable.	Negative
VAL-10	Error message clarity	Trigger validation/error conditions.	Messages are understandable and useful.	Positive

4. Potential Bugs / Risk Areas

The following items are potential risks identified through test analysis without executing the application.

ID	Area	Potential Bug	Severity	Reason / Impact
BUG-01	Registration	Duplicate email allowed during registration	Major	May create duplicate accounts and data inconsistency.
BUG-02	Authorization	Unauthenticated user can access protected task data	Critical	Can expose private user information and bypass security controls.
BUG-03	Authorization	One user can view or modify another user's task	Critical	Serious privacy and data integrity risk.
BUG-04	Delete / Persistence	Deleted task reappears after refresh	Major	May indicate deletion is not correctly persisted in the database.
BUG-05	Create / Validation	Empty or spaces-only task title is accepted	Major	Invalid task records may be stored.
BUG-06	Validation	Invalid input causes application/server error	Major	Can affect stability and user experience.
BUG-07	Edit / Persistence	Updated task changes are lost after refresh	Major	Users may lose changes due to persistence failure.
BUG-08	Login	Incorrect credentials are accepted	Critical	Could allow unauthorized account access.
BUG-09	UI / Validation	Extremely long input breaks layout or UI	Minor	Affects usability and interface stability.
BUG-10	Error Handling	Technical or database details exposed in error messages	Major	May reveal implementation details useful to attackers.

5. Testing Priorities

Priority	Focus
High / Critical	Authentication, authorization, unauthorized task access, login security, data privacy.
Medium / Major	Task CRUD functionality, data persistence, validation, error handling.
Lower / Minor	UI layout, usability issues, non-blocking boundary presentation issues.

6. Testing Approach

- Start with registration and login to verify authentication flow.
- Test Create, View, Edit, and Delete operations using valid data.
- Add negative cases for missing, invalid, or unauthorized input.
- Test edge and boundary conditions such as spaces-only and extremely long input.
- Verify persistence by refreshing after create, edit, and delete operations.
- Check authorization by attempting access to another user's task.
- Review error behavior for invalid requests and server/database failures.

7. Assumptions

- Only logged-in users should be able to manage tasks.
- Users should only access and modify tasks they are authorized to manage.
- Required fields should reject empty and spaces-only values.
- Task data should persist after successful create or update.
- Successfully deleted tasks should not reappear.
- Error messages should be clear and should not expose unnecessary technical details.

8. Conclusion

This assessment provides structured test coverage for Registration, Login, Task CRUD operations, Input Validation, and Error Handling. The highest release priorities are authentication, authorization, data privacy, data integrity, and reliable persistence of create, update, and delete operations. The listed bugs are potential risks that require verification through actual application testing.